Jordan B. S. Van Beeck

Asteroseismology | Stellar Astrophysics | FOSS

Personal Webpage Jordan-Van-Beeck jordan-van-beeck Google Scholar
JVB11 ArXiv ADS library GSI-5206-2022 0000-0002-5082-3887 Zenodo

Scientific Education

Ph.D., Astronomy & Astrophysics, KU Leuven (Belgium), September 2023

Asteroseismology of Kepler B stars: internal magnetism and non-linear mode coupling

Promoters: Prof. Dr. C. Aerts, Prof. Dr. T. Van Hoolst, Prof. Dr. D. M. Bowman (thesis download link)

M.Sc., Astronomy & Astrophysics, KU Leuven (Belgium), July 2019 – *Magna Cum Laude*

The influence of an interior magnetic field on gravity-mode oscillations of intermediate-mass stars

Promoters: Prof. Dr. C. Aerts, Dr. T. Van Reeth, Prof. Dr. D. M. Bowman (thesis download link)

M.Sc., Chemistry, University of Antwerp (Belgium), September 2017 – *Magna Cum Laude*

Characterization of radioactive particles

Promoters: Prof. Dr. K. Janssens, Prof. Dr. B. Salbu, Prof. Dr. O.-C. Lind, M.Sc. G. Nuyts

B.Sc., Chemistry, University of Antwerp (Belgium), September 2015 – *Magna Cum Laude*

Atomic scale reactive MD studies of DNA oxidation for plasma oncology: the role of H_2O_2 and HO_2

Promoters: Prof. Dr. A. Bogaerts, Prof. Dr. E. Neyts, Dr. C. Verlackt

Employment

Heidelberg Institute for Theoretical Studies (HITS), TOS group, Heidelberg, Germany

Postdoctoral Researcher Asteroseismology February 2025 – February 2027

Project: higher-order rotational effects on mixed-mode oscillations of red giants

KU Leuven, Department of Physics and Astronomy, Institute of Astronomy, Leuven, Belgium

Research Associate, Asteroseismology, September 2023 – December 2023

Project: non-linear asteroseismic modeling of slowly pulsating B stars

Awards and Fellowships

2021 - 2022	FWO (Research Foundation Flanders) long research stay grant **(€12 738)** for a stay as a VSR at TAPIR, Caltech, CA, USA with Prof. Dr. J. Fuller.
2019 - 2023	4-year PhD Fellowship, Department of Physics & Astronomy, KU Leuven
2019	Paul Smeyers Prize, KU Leuven Awarded to the annual best master's thesis in Astronomy & Astrophysics at the June examination session.

Teaching Experience

Lecturer, Heidelberg University

2025 Theoretical introduction for 'Asteroseismology'
(Master-level physics course that introduces various aspects of asteroseismology.)

Teaching Assistant, Department of Physics & Astronomy, KU Leuven

2022 - 2023 K01B4A: 'Natuurkunde met elementen van wiskunde' I (taught in Dutch)
(Introductory physics course for the first-year students in the B.Sc. of pharmaceutical sciences)

2019 - 2021 K01B4A|B:'Natuurkunde met elementen van wiskunde' I|II (taught in Dutch)
English translation of course title: 'Physics with elements of mathematics' I|II
(Introductory physics courses for the first-year students in the B.Sc. of pharmaceutical sciences)

Visiting Research Positions (abridged)

11/2021 - 06/2022	*Visiting Student Researcher (VSR) at TAPIR, Caltech, CA, USA* Project goal: development of computational tools that aid in explaining oscillation mode amplitude limitation in slowly pulsating B stars Mentor: Prof. Dr. J. Fuller — *Funding: FWO long research stay grant*
11/2021 - 12/2021	*Transport in Stellar Interiors Program Affiliate (workshop, conference) at KITP, UC Santa Barbara, CA, USA* Mentor: Prof. Dr. L. Bildsten — *Funding: FWO long research stay grant*

Seminars/Conference Presentations (abridged)

July 2025	CONTRIBUTED TALK (doi:10.5281/zenodo.17241657) TASC9 / KASC16 workshop/conference at ISTA, Austria.
March 2023	GOOD VIBRATIONS SEMINAR (https://youtu.be/dWaEkri_s2I?feature=shared) The Good Vibrations seminar series, S3E8.
February 2023	AD-VALVAS SEMINAR (https://youtu.be/nTWZoLCMa_g?feature=shared) Ad-valvas seminar of the Institute of Astronomy, KU Leuven, Belgium.
July 2022	POSTER PRESENTATION (additional info: doi:10.5281/zenodo.6814504) TASC6 / KASC13 workshop/conference at KU Leuven, Belgium.
November 2021	CONTRIBUTED TALK (doi:10.26081/K6VH15) Transfer in Stellar Interiors conference (part of the Probes of Transport in Stars program), KITP, UC Santa Barbara, CA, USA.
July 2020	CONTRIBUTED TALK (doi:10.5281/zenodo.5525346) MOBSTER-1 Virtual conference 2020: Stellar variability as a probe of magnetic fields in massive stars (online, hosted by University of Delaware, DE, USA)

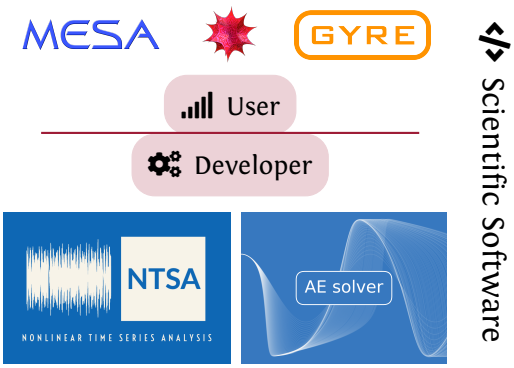
Professional Activities

Conference organization

July 2022 Member of the Local Organizing Committee
TASC6 / KASC13 workshop/conference in Leuven, Belgium

Professionally associated as

Fellow (Graduate Student)
2022 *Royal Astronomical Society (RAS)*

Graduate Student Member
2020 - 2022 *American Astronomical Society (AAS)*
2020 - 2022 *Royal Netherlands Astronomical Society (KNA)*
2019 - 2023 *International Research Network for Nuclear Astrophysics (IReNA)*

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